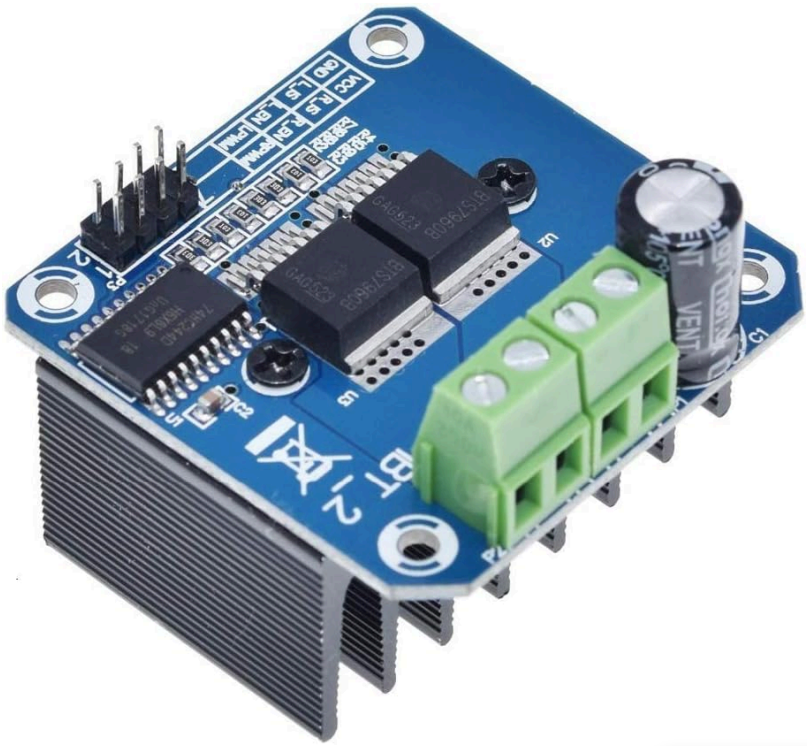
BTS7960 Motor Driver Datasheet and Circuit Diagram

Update Time: Oct 11, 2023 Readership: 193209



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and an n-channel low-side MOSFET, together with an integrated driver IC, all compactly packaged as a single unit.

What sets the BTS7960 apart is its clever deployment of a p-channel high-side switch, eliminating the need for a charge pump and effectively reducing electromagnetic interference (EMI). This module boasts seamless compatibility with microcontrollers, courtesy of the integrated driver IC. This versatile IC is loaded with features, including logic-level inputs, current sensing diagnostics, slew rate adjustment, dead time control, and robust protection mechanisms against overtemperature, overvoltage, undervoltage, overcurrent, and short circuits.

In essence, the BTS7960 serves as a cost-effective solution, meticulously optimized to safeguard high-current PWM motor control. All of this is achieved while keeping a watchful eye on the conservation of valuable board space.

Features

Feature	Description
Low Path Resistance	16 mΩ at 25 °C

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Robust Current Limit	43 A typical
Diagnostic Capabilities	Status flag diagnosis with current sensing
Temperature & Voltage Safeguards	Over Temperature shutdown, overvoltage lockout, undervoltage shutdown
Logic-Level Inputs	Driver circuit with logic-level inputs
EMI Optimization	Adjustable slew rates for optimal EMI performance

Description

The BTS7960 is a vital component of the NovalithIC family, bringing together three distinct elements packaged as a unified powerhouse. Inside this compact module, you'll find a p-channel high-side MOSFET, an n-channel low-side MOSFET, and a dedicated driver IC, all seamlessly integrated to create a high-current half-bridge solution. What's truly innovative is their

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is particularly advantageous. Thanks to the p-channel high-side switch, the requirement for a charge pump is elegantly eliminated, effectively quelling electromagnetic interference (EMI). Furthermore, interfacing with a microcontroller is a breeze, courtesy of the integrated driver IC. It offers an array of features, including logic-level inputs, current sensing diagnostics, customizable slew rate adjustments, and a robust array of safeguards covering overtemperature, overvoltage, undervoltage, overcurrent, and short circuit protection.

But it doesn't stop there; the BTS7960 is highly adaptable. It can be paired with additional BTS7960 units to create versatile H-bridge and 3-phase drive configurations, allowing for a wide range of motor control applications.

Maximum Ratings

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3.0.3	HS/LS continuous drain current	$I_{D(HS)}$ $I_{D(LS)}$	-40	40[1)]	A	T_C < 85°C switch active
3.0.4	HS pulsed drain current	$I_{D(HS)}$	-60	60[1)]	A	T_C < 85°C t_{pulse} = 10ms
3.0.5	LS pulsed drain current	$I_{D(LS)}$	-60	60[1)]	A	
3.0.6	Voltage at SR pin	V_{SR}	-0.3	1.0	V	
3.0.7	Voltage between VS and IS pin	V_{VS} -V_{IS}	-0.3	45	V	
3.0.8	Voltage at IS pin	V_{IS}	-20	45	V	
Thermal Maximum Ratings						
3.0.9	Junction temperature	T_j	-40	150	°C	
3.0.10	Storage temperature	T_{stg}	-55	150	°C	
ESD Susceptibility						
3.0.11	ESD susceptibility HBM IN, INH, SR, IS OUT, GND, VS	V_{ESD}	 -2 -6	 2 6	kV	according to EIA/ JESD 22-A 114B

[1)] Maximum reachable current may be smaller depending on current limitation level

Application

Application	Examples	Benefits
Automotive Systems	Power windows, sunroofs,	Efficient and reliable motor control
	power seats, windshield	
	wipers, HVAC systems, etc.	

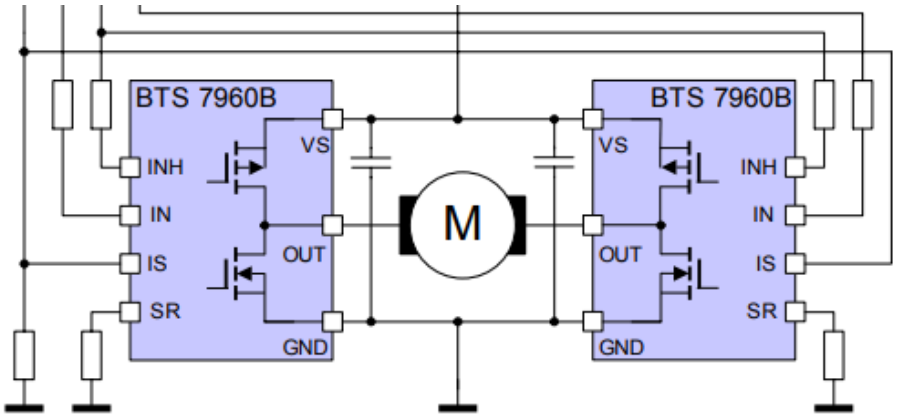
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Industrial Automation	Conveyor systems, material	Reliable motor control in industrial
	handling equipment,	automation, enhancing productivity.
	manufacturing machinery,	
	robotic assembly lines, etc.	
Consumer Electronics	Camera gimbals,	Stable and accurate motor control in
	camera stabilizers,	consumer electronics for smooth camera
	motorized camera sliders,	movement.
	etc.	
Model Cars and Drones	Remote-controlled model	Precise speed and direction control for

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Battery-Powered Devices		
	electric scooters, e-bikes,	efficiency in battery-powered devices.
	portable devices, etc.	
Home Automation	Window blinds, garage door	Motorized control of home automation
	openers, smart curtains,	devices for convenience and efficiency.
	etc.	
Educational Projects	DIY electronics projects,	Learning motor control concepts and
	student projects, etc.	creating practical electronic systems.

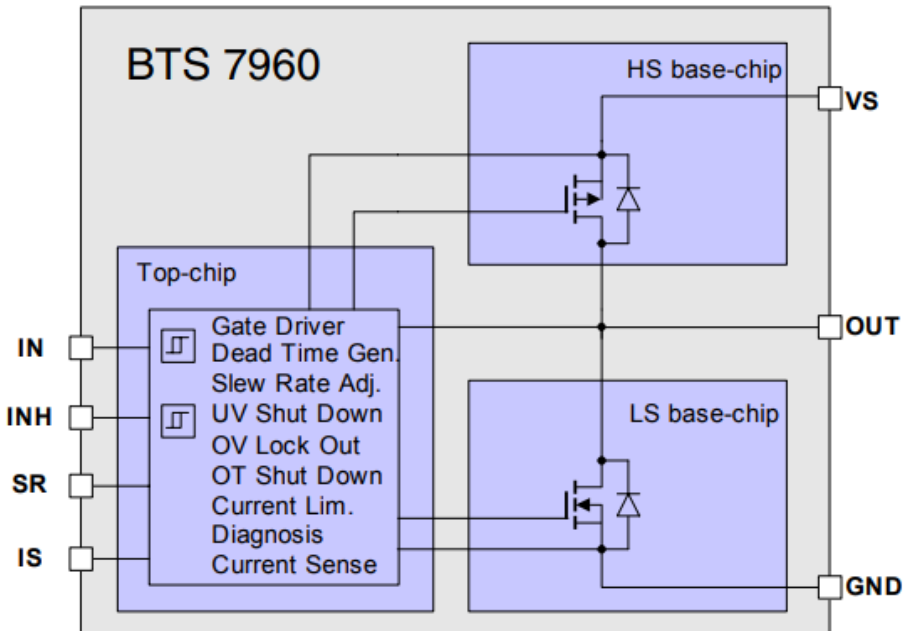
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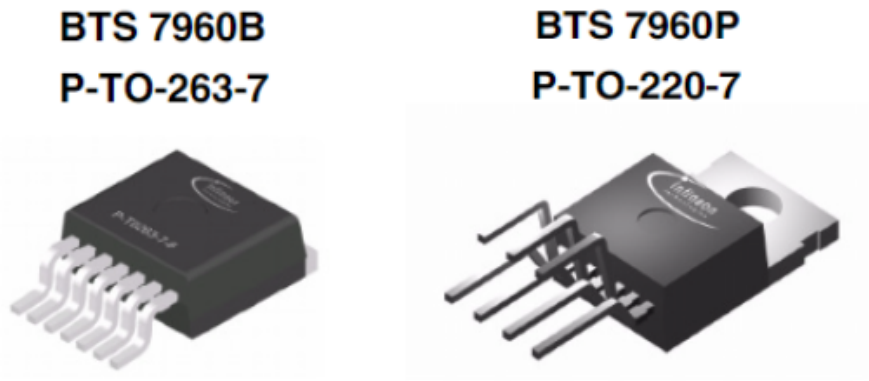
High Current H-Bridge

Application Example: H-Bridge with two BTS7960B

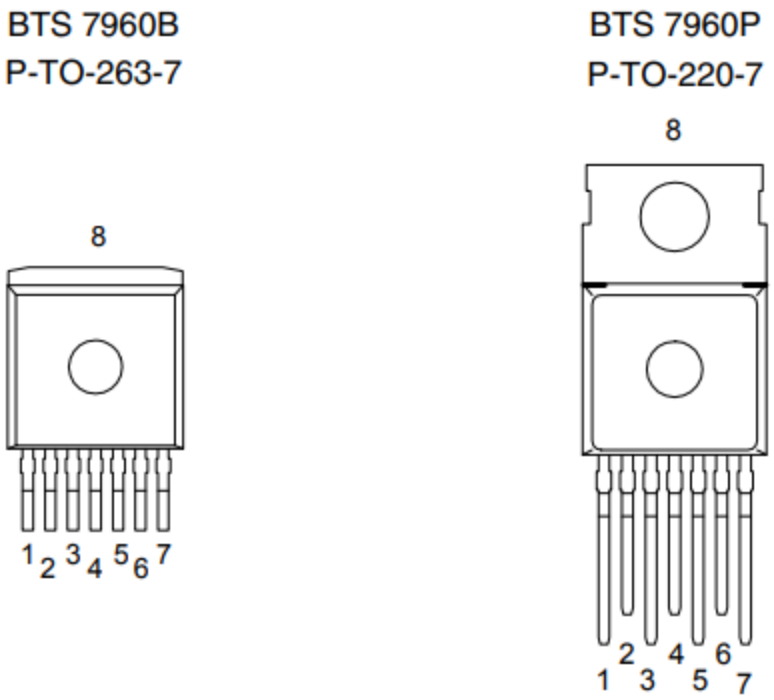
Block Diagram



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BTS7960B & BTS7960P



Pin Assignment

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5	SR	I	Slew Rate The slew rate of the power switches can be adjusted by connecting a resistor between SR and GND
6	IS	O	Current Sense and Diagnosis
7	**VS**	**-**	**Supply**

Pin Definitions and Functions

Control and Diagnostics

- Input Circuit

The BTS7960's control inputs, IN and INH, are equipped with TTL/CMOS-compatible Schmitt triggers featuring hysteresis. These controls manage the integrated gate drivers for the MOSFETs. To enable the device, set the INH pin to a high state, and this allows one of the power switches to turn on based on the status of the IN pin. For deactivation of both switches, simply set the INH pin to a low state. Notably, no external driver is needed as the BTS7960 seamlessly interfaces with a microcontroller.

- Dead Time Generation

In bridge configurations, it's paramount to prevent simultaneous conduction of the high-side and low-side MOSFETs, avoiding a direct connection of battery voltage to ground. The driver IC

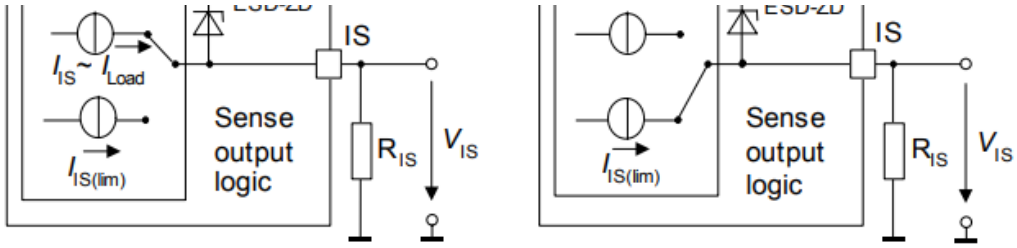
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To fine-tune electromagnetic emissions and strike a balance between power dissipation, the switching speed of the MOSFETs can be customized using an external resistor. The slew rate pin, SR, empowers users to optimize the equilibrium between emissions and power dissipation within their unique applications. This can be achieved by connecting an external resistor, RSR, to ground.

- Status Flag Diagnosis With Current Sense Capability

The status pin, IS, assumes a dual role, functioning as both a current sense and error flag output. During normal operation in current sense mode, a current source is linked to the status pin, delivering a current that is proportionate to the forward load current flowing through the active high-side switch. In instances where the high-side switch is inactive or current flows in reverse, no current is driven except for a marginal leakage current, IIS(LK). The voltage per output current is determined by the external resistor, RIS. For example, with a nominal current sense ratio of 8500 (kILIS = IL / IIS), a 1kΩ resistor results in VIS = (IL / 8.5 A)V. In the event of a fault condition, the status output is connected to a current source that operates independently of the load current, providing IIS(lim). The maximum voltage at the IS pin is dependent on the choice of the external resistor and the supply voltage. When it comes to current limitation scenarios

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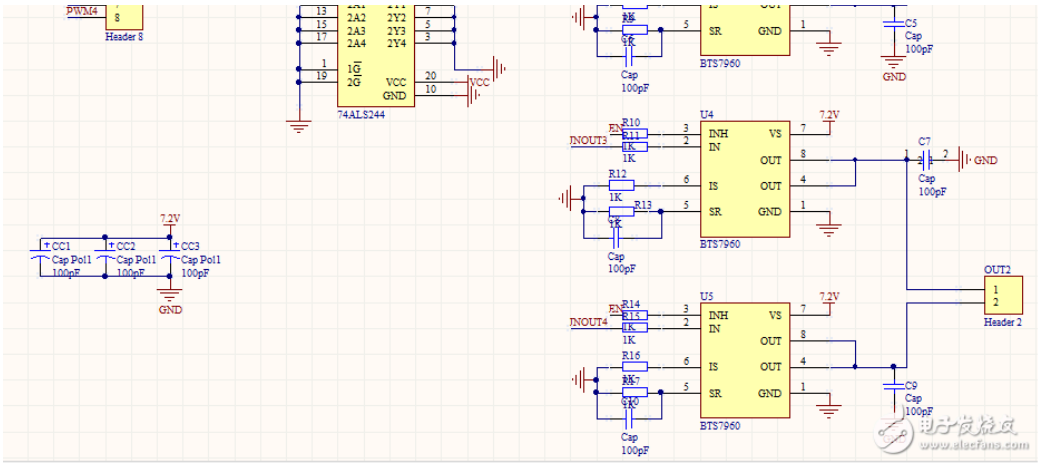


Sense current and fault current

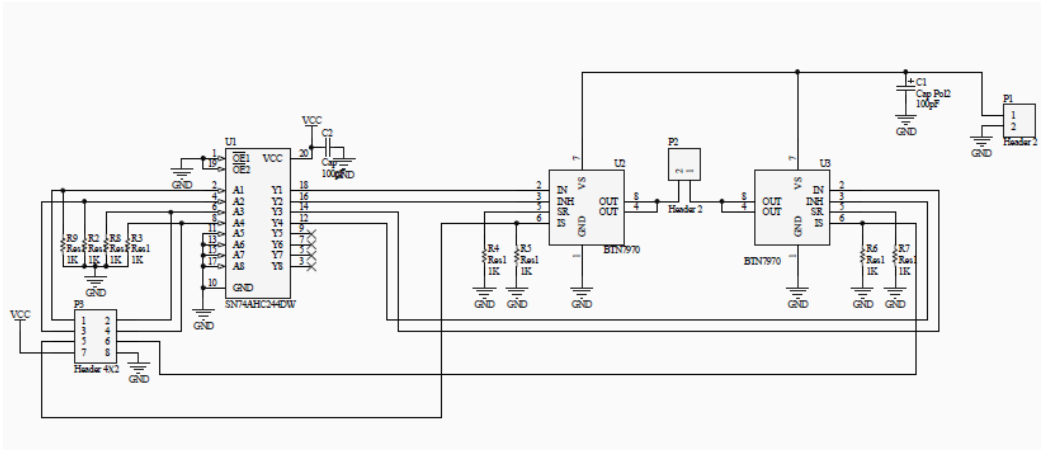
BTS7960 Circuit Diagram

This is a potential motor drive circuit that may be used in electronic competitions. It includes the PCB and schematic of the BTS7960 dual motor driver board. The circuit uses the BTS7960 as the motor driver chip and incorporates the 74HC244PW for isolation (utilizing the TSSOP package). Additionally, the design takes into account screw holes that can be directly mounted on the existing screws of the C or D car model motor seat. The power lines are of ample width, ensuring effective heat dissipation.

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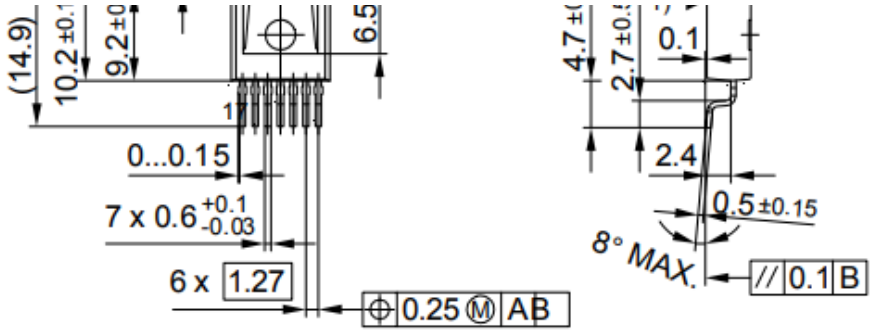


BTS7960 Module Schematic



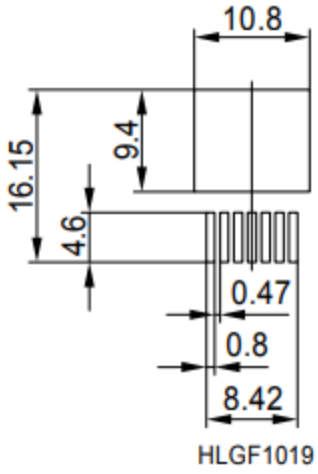
BTS7960 Dimensions

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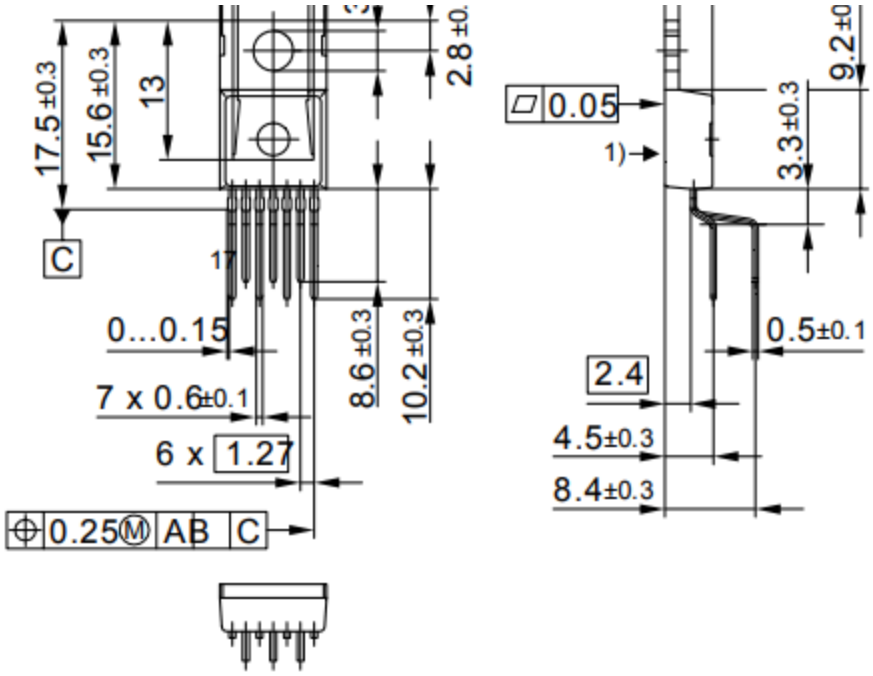
1) Shear and punch direction no burrs this surface
— Back side, heatsink contour
All metal sufaces tin plated, except area of cut .

P-TO-263-7 (Plastic Transistor Single Outline Package)



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— Back side, heatsink contour
All metal surfaces tin plated, except area of cut.

P-TO-220-7 (Plastic Transistor Single Outline Package)

BTS7960 Manufacturer

The BTS7960 is a motor driver integrated circuit manufactured by Infineon Technologies. Infineon Technologies is a global semiconductor manufacturing company headquartered in

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mobility, and security. Infineon is known for its innovations in power management, security, and connectivity technologies, making a significant impact on the electronics industry.

Popular YouTube Videos



How to work with BTS7960 motor driver and Arduino

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Extended Reading

- BTS7960 Motor Driver Datasheet and Circuit Diagram
- IRLZ44N MOSFET Datasheet, Circuit, Equivalent, Pinout
- 7408 Integrated Circuit Datasheet: Pinout, Pin Diagram, Truth Table
- 7404 Integrated Circuit (IC): Datasheet, Pinout, Pin Diagram, Truth Table

FAQ

What is the PWM frequency of BTS7960

25 kHz

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What is the use of BTS7960 motor driver

The BTS7960 motor driver is designed to efficiently manage the operation of a single motor. It necessitates a power source within the voltage range of 6-27V DC, a dedicated +5V logic power supply, and one or two +5V PWM inputs for precise control over both the motor's speed and direction. The BTS7960 offers the flexibility to function in three distinct wiring configurations, catering to diverse application requirements.

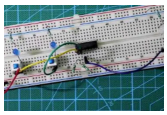
What is the equivalent of BTS7960

BTN8982TA

Stella Brinkley

Stella Brinkley is a senior electronics engineer with 6 years of experience, specializing in the detailed study of resistor, transistor and package design. Her comprehensive knowledge allows her to drive innovation and excellence in the electronics industry.

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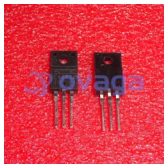
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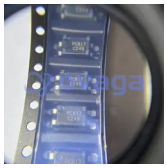
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